

AdSpotX

Integrated AdSpot + Partner Network Platform

Document date: 4 August 2026

Confidential - AdSpot Nigeria / Africa

Architecture - Installation - CI/CD - Personas - Market Analysis - Revenue Forecasts - Partner Flow

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Document date: 4 August 2026

Version: 1.2.0 (AdSpotX integrated release)

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1. Application overview

What is AdSpotX?

AdSpotX is the branded, integrated network partner product inside the AdSpot unified platform. It combines:

- The core AdSpot rewarded-attention stack (reviewers, brands, admin)
- The modular partner-portal for newspapers, digital publishers, and media outlets
- Admin tooling to onboard publishers, activate AdSpot routing, and view network analytics

AdSpotX lets media partners monetise editorial inventory with verified brand-review campaigns without building ad tech. Partners embed a one-click Integrate with AdSpot flow; campaigns route only after explicit opt-in (adspot_linked = true).

Value proposition

Stakeholder	Benefit
Publishers / partners	Rev-share on verified completions; one-click integration; no upfront ad-tech build
Brands	Reach engaged audiences via short-form video reviews on trusted media properties
Reviewers	Earn points on the existing earn portal (/earn) for authentic feedback
AdSpot platform	Network distribution, platform fee, fraud-scored completions, first-party survey data

Commercial model (pilot)

- Rev-share: 60% publisher / 30% reviewer pool / 10% AdSpot platform (negotiable at scale)
- Verification: Completion webhooks + minimum watch-time before payout
- Pilot: 90-day trial; integration inactive until partner opts in
- Settlement: Monthly NGN statements; Paystack/Flutterwave rails for brand top-ups

2. Technical architecture

Monorepo structure

Path	Package	Purpose
app/	@workspace/adspot-unified	Unified React SPA (landing + earn + brands + partners mount)

partner-portal/	@workspace/partner-portal	Standalone partner module (dashboard, slots, revenue, integrate button)
server/	@workspace/api-server	Express REST API
lib/db/	@workspace/db	Drizzle schema, migrations, seeds
lib/api-zod/	@workspace/api-zod	Zod API contracts
lib/api-client/	@workspace/api-client	Typed frontend client
scripts/	-	Hostile audit, packaging, PDF render

Build order: api-server -> adspot-unified -> partner-portal

Unified start: PORT=3001 STATIC_DIR=./app/dist node server/dist/index.mjs

System architecture

[illegible]

API routes & auth

Router	Prefix	Key endpoints
auth	/api/auth	register, login, me
ads	/api/ads	Feed, ad detail (video creatives)
reviews	/api/reviews	start, complete sessions
points	/api/points	balance, ledger
brands	/api/brands	analytics, comments, AI summary
admin	/api/admin	users, events, stats, redemptions
partners	/api/partners	list, create, integration activate/deactivate
health	/api/healthz	DB connectivity

Roles

Role	Portal	Capabilities
reviewer	/earn/login	Watch ads, earn points, leaderboard
brand	/brands/login	Campaigns, analytics, AI summaries
admin	/brands/login -> /brands/admin/*	Users, events, financials, AdSpotX
super_admin	/brands/login	All admin capabilities + owner bypass

Hierarchy: super_admin bypasses all role checks; admin satisfies admin routes. Demo password: password123.

AdSpotX partner integration

Method	Path	Description
GET	/api/partners	List partners (admin auth)
POST	/api/partners	Create partner + inactive integration
GET	/api/partners/:id/integration	Status (default inactive)
POST	/api/partners/:id/integration/activate	Flip adspot_linked=true, issue API key + embed tag
POST	/api/partners/:id/integration/deactivate	Deactivate

Environment: AUDIT_PARTNER MOCK=1 enables in-memory partner + reviewer mock data for audit/screenshots without Postgres.

3. Installation guides

Prerequisites (all targets)

- Node.js 20+
- pnpm 9 (corepack enable && corepack prepare pnpm@9 --activate)
- PostgreSQL 14+ (RDS, Railway Postgres, or local)

Local / engineer quick start

```
cd AdSpot-Unified-3
npx pnpm@9 install
cp server/.env.example server/.env # set DATABASE_URL, JWT_SECRET
npx pnpm@9 run build
npx pnpm@9 --filter @workspace/db run push
npx pnpm@9 --filter @workspace/db run seed:accounts

PORT=3001 STATIC_DIR=./app/dist node server/dist/index.mjs
# Mock mode (no DB): AUDIT_PARTNER MOCK=1
```

Open <http://localhost:3001> - landing, /earn, /brands/login, /partners on one origin.

AWS deployment (expanded)

Option A - Unified deploy (recommended)

Single Node process serves SPA + partner portal + API.

AWS App Runner

1. Connect repo or push container image.
2. Start command: ``node server/dist/index.mjs``
3. Environment:

Variable	Value
PORT	8080
DATABASE_URL	RDS Postgres connection string
JWT_SECRET	openssl rand -hex 64
STATIC_DIR	./app/dist
ADSPOT_PUBLIC_URL	https://your-domain.com
CORS_ORIGINS	Empty for same-origin

4. Run migrations (one-off task or CI):

```
npx pnpm@9 --filter @workspace/db run push
npx pnpm@9 --filter @workspace/db run seed:accounts
```

ECS Fargate / Elastic Beanstalk: Same env vars; ship app/dist + server/dist in container.

Option B - Split deploy

Component	AWS service	Notes
Frontend SPA	S3 + CloudFront	Deploy app/dist/; 403/404 -> /index.html
Partner portal	Embedded at /partners or partner-portal/dist on second CloudFront	
API	App Runner / ECS / ALB	server/dist/index.mjs
Database	RDS PostgreSQL 14+	Private subnet; SG allows 5432 from compute only

Build split frontend:

```
VITE_API_BASE_URL=https://api.yourdomain.com npx pnpm@9 run build
```

Set on API: CORS_ORIGINS=https://app.yourdomain.com

RDS setup

1. Create PostgreSQL 14+ in private subnet.
2. Security group: inbound 5432 from App Runner / ECS only.
3. ``DATABASE_URL=postgresql://user:pass@host:5432/adspot``
4. Migrate and seed (see above).
5. Verify ``/api/healthz`` returns 200 with ``"db":"connected"``.

Post-deploy checklist

- [] pnpm build passes
- [] /api/healthz -> 200
- [] /earn/login, /brands/login, /partners -> 200
- [] Partner integration inactive by default; activate returns apiKey

- [] node scripts/hostile-audit.mjs -> PASS or PARTIAL
- [] TLS valid; rotate demo passwords before production

Railway deployment

1. **Create project** - connect GitHub repo `AdSpot-Unified-3`.
2. **Add Postgres** - Railway Postgres plugin; copy `DATABASE\_URL` into service variables.
3. **Service settings:**

Variable	Value
PORT	\${PORT} (Railway injects)
DATABASE_URL	From Postgres plugin
JWT_SECRET	Random 64-byte hex
STATIC_DIR	/app/dist
ADSPOT_PUBLIC_URL	https://\${RAILWAY_PUBLIC_DOMAIN}
NODE_VERSION	20

4. **Build command:**

```
npx pnpm@9 install && npx pnpm@9 run build
```

5. **Start command:**

```
node server/dist/index.mjs
```

6. **Release command** (optional, run on deploy):

```
npx pnpm@9 --filter @workspace/db run push && npx pnpm@9 --filter @workspace/db run seed:accounts
```

7. Enable **public networking**; Railway auto-deploys on git push to connected branch.

4. Cursor -> hosting CI/CD

Practical pipeline: edit -> commit -> push -> deploy

```
Cursor IDE (local edits)
-> git commit + push to GitHub
-> CI (.github/workflows/ci.yml): typecheck, lint, test, build
-> Hosting auto-deploy (Railway / AWS)
-> /api/healthz smoke + hostile-audit (optional)
```

GitHub Actions (included)

.github/workflows/ci.yml runs on every push/PR:

- pnpm install --frozen-lockfile
- pnpm typecheck
- pnpm lint / pnpm test
- pnpm build

Extend with deploy job:

```
# Example: Railway deploy after CI (add RAILWAY_TOKEN secret)
deploy:
  needs: verify
  runs-on: ubuntu-latest
  if: github.ref == 'refs/heads/main'
  steps:
    - uses: actions/checkout@v4
    - run: npx railway up --service adspot-api
```

AWS deploy from Cursor

- 1. Push to `main` -> CodePipeline / App Runner auto-build from repo.
- 2. Or: Cursor Cloud Agent completes feature -> PR merge -> ECS/App Runner picks up image.
- 3. Use `scripts/package-staging.mjs` for zip handoff to EC2 if needed.

Cursor Cloud Agents workflow

- 1. Open project in Cursor; Cloud Agent implements change on branch.
- 2. Agent runs `pnpm build` + `node scripts/hostile-audit.mjs --mock-only`.
- 3. Push branch; open PR; CI verifies.
- 4. Merge -> Railway/AWS auto-deploy.

Demo credentials (staging)

Role	Email	Password
Super admin	oadeagbo@gmail.com	password123
Admin	admin@adspot.demo	password123
Reviewer	alice@reviewer.demo	password123
Brand	brand@adspot.demo	password123

5. Features & capabilities by persona

Reviewers (earn portal - `/earn`)

Feature	Route / component
Landing hero Start earning	/ - landing-start-earning-hero
Dashboard, points, ad feed	/earn/dashboard
Campaign review + video	/earn/review/:id - VideoPlayer, review-video-player
Leaderboard	/earn/leaderboard
Earnings / ledger	Points API, collapsible history
Registration	/earn/register

Brands (`/brands`)

Feature	Route
Login / register	/brands/login, /brands/register
Campaign analytics dashboard	/brands/dashboard - charts, demographics, filters
AI analytics summary	Generate AI Summary on dashboard
Campaign CRUD	/brands/ads, /brands/ads/new
Settings	/brands/settings

Admin (`/brands/admin`)

Feature	Route
Overview	/brands/admin/dashboard
User directory	/brands/admin/users
Event log	/brands/admin/events
Financials (KPIs, redemptions, ledger)	/brands/admin/financials
AdSpotX partner management	/brands/admin/adspotx
Ad moderation	/brands/admin/ads

Partners (AdSpotX portal - `/partners`)

Feature	Route
Partner dashboard	/partners
Integration + Integrate with AdSpot	/partners/integration
Inventory slots	/partners/slots
Revenue view	/partners/revenue
Network directory	/partners/partners

6. Application screenshots

Captured from live unified server (PORT=3004, AUDIT_PARTNER MOCK=1, August 2026).

Landing - Start earning hero

A

AdSpot

Benefits

How it works

Pricing

Start earning

Sign in

Brand

Watch ads. Share feedback. Get paid.

AdSpot connects Nigerian brands with real people. Reviewers earn rewards. Brands get verified engagement and honest answers.

Start earning

Already have an account? [Sign in](#) - [Register as brand](#)

Sign in to your account

Reviewer

Watch ads, share feedback, earn points.

Sign in →

Brand

Run campaigns and see real results.

Sign in →

Admin

Manage users, ads, and payouts.

Sign in →

See how it works

Sample advert — reviewers watch, answer questions, and earn points.

Built for reviewers and brands

Whether you want to earn or advertise, AdSpot gives you what you need.

Landing page with Start earning CTA

Reviewer dashboard

A

ADSPOT

A

alice

REVIEWER

DASHBOARD

LEADERBOARD

ADSPOT > REVIEW ADS

YOUR POINTS

2,840

pts

Start reviewing →

Available ads

RICK ASTLEY

NEVER GONNA GIVE YOU UP

4 K

DEMO BRAND

Summer Splash Campaign

★ +120 pts

Review →

This week's top reviewers

1

You

2,840 pts

Points history

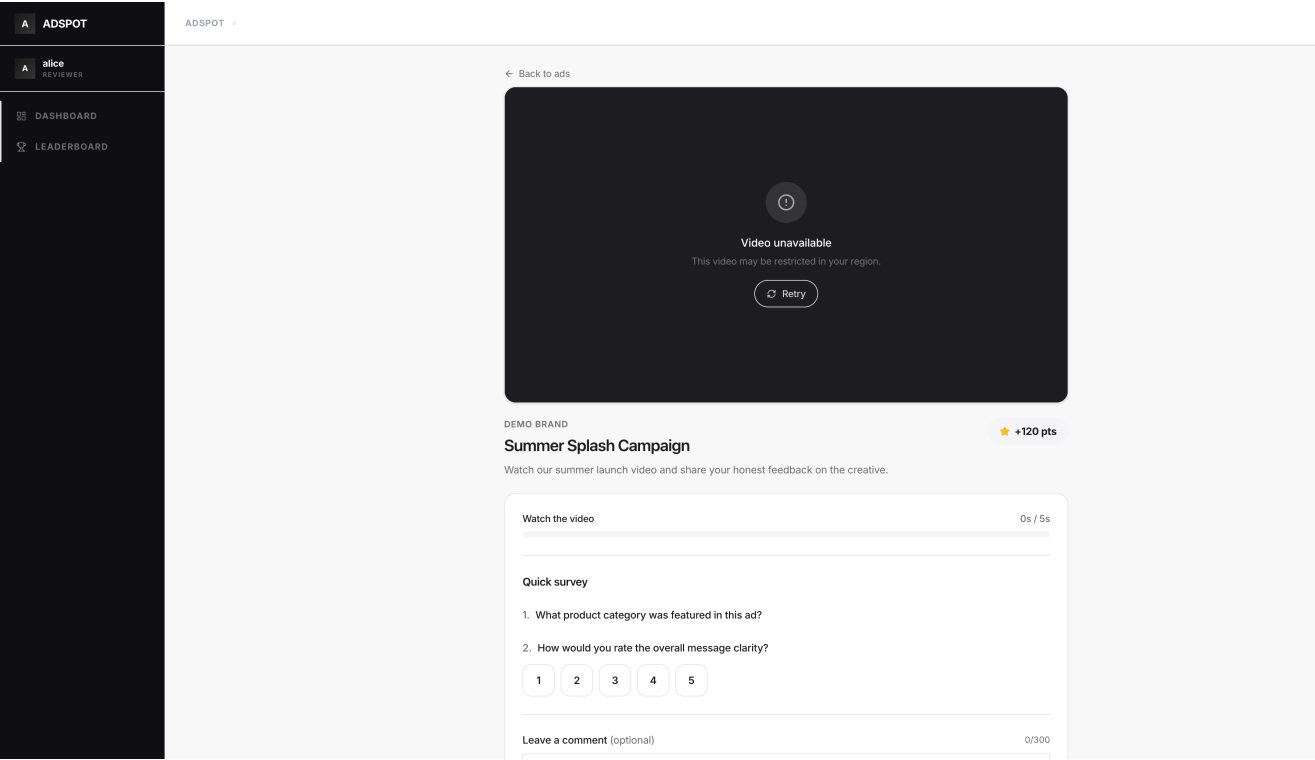
Show

Reviewer dashboard with points and available ads

Reviewer campaign review (video + survey)

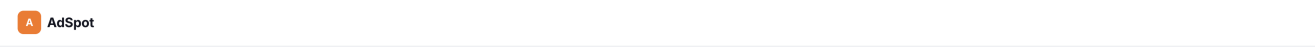
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Campaign review session with video player and questions

Brand login



Brand or admin sign in

Use the same login for brand and admin accounts.

Looking to earn as a reviewer?

Reviewer

Email

you@company.com

Password

Your password

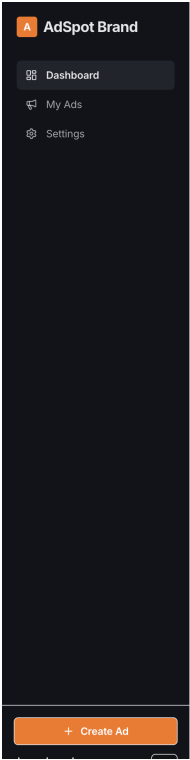
Sign in

No account? Register

Back to home

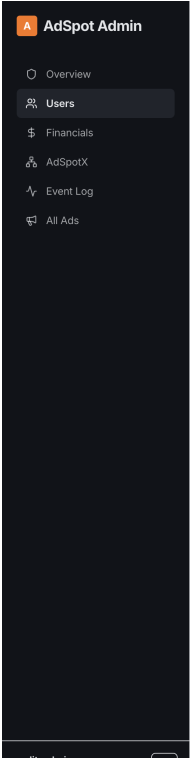
Brand and admin sign-in portal

Brand analytics dashboard



Brand campaign analytics with charts and KPIs

Admin - Users



Admin user directory with roles

Admin - Event log

Campaign Analytics

Real-time performance insights for your Nigerian audience

Filters:

All Campaigns

All Genders

All Ages

All States

Any Time

Total Views

12,480

Completions

8,920

Completion Rate

71.0%

Avg Watch Time

42s

Points Spent

1,068,000

total cost in pts

Engagement

undefined%

completion rate

14-Day Completion Trend

Daily completions over the past 2 weeks



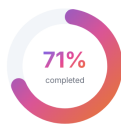
Gender Split

0 profiled reviewers

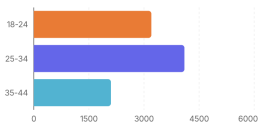


Completion Rate

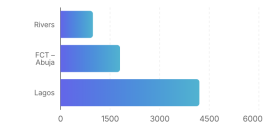
Share of views that finish



Age Bands

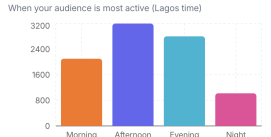


Top States



Time of Day

When your audience is most active (Lagos time)



Audience Deep-Dive

Filter by any demographic to see who is actually completing your campaigns.

User Directory

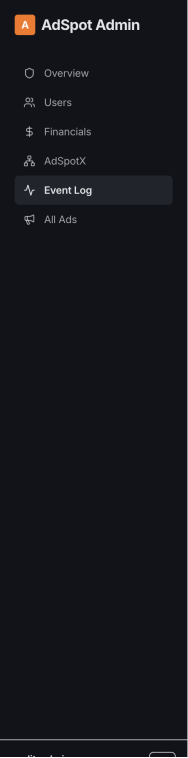
Manage all platform accounts

All Roles

User	Email	Role	Balance	Joined
platform-owner	oadeagbo@gmail.com	Super Admin	-	4/6/2026
audit-admin	admin@adspot.demo	Admin	-	5/6/2026
alice	alice@reviewer.demo	Reviewer	2,840	6/5/2026
demo-brand	brand@adspot.demo	Brand	-	6/20/2026

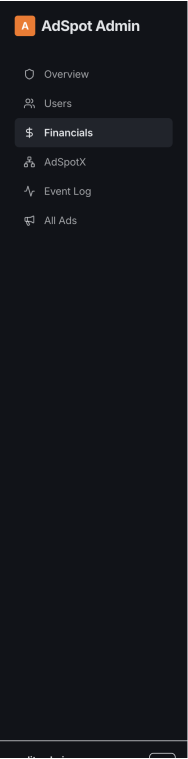
Showing 4 of 4 users

PreviousNext



Admin platform event audit trail

Admin - Financials



Admin financial activity and redemptions

Event Log

Audit trail of all platform activity

Event Type
e.g. ad_created

From Date
dd/mm/yyyy

To Date
dd/mm/yyyy

Filter

Clear

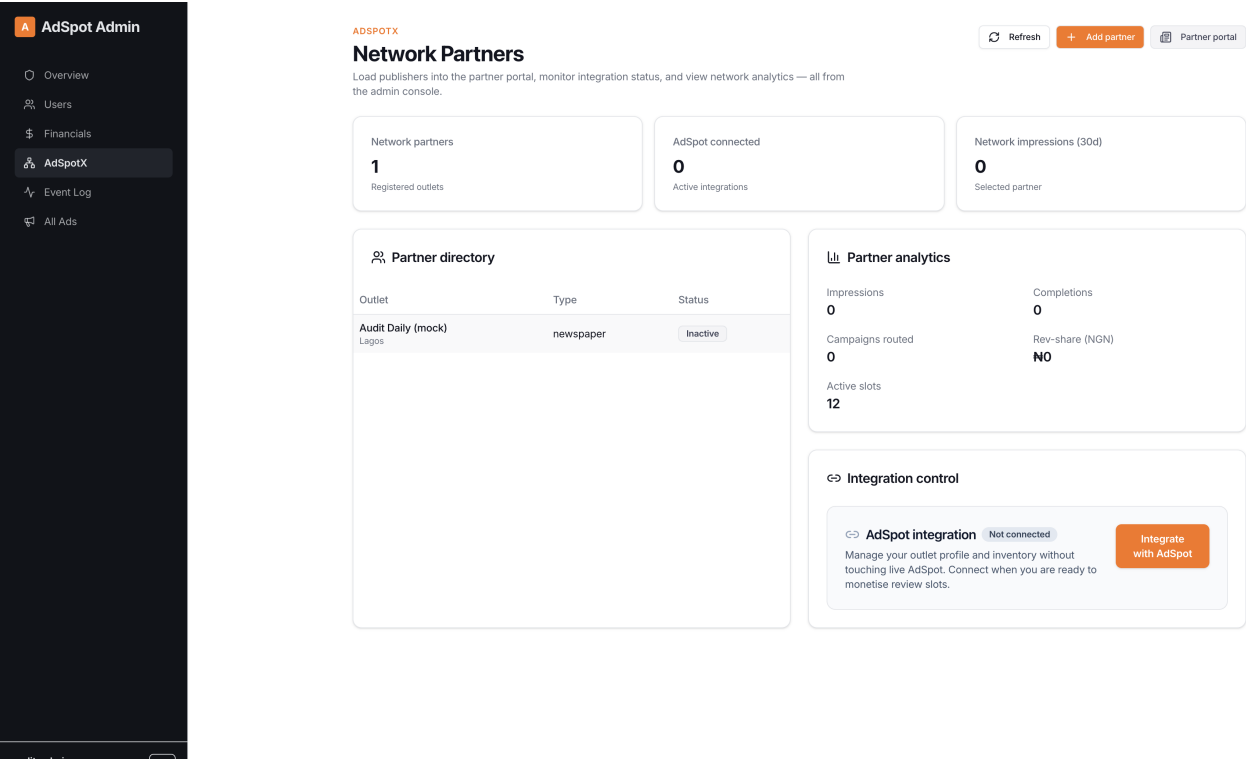
Time	Event Type	Actor	Entity	Metadata
8/4/2026, 5:01:09 AM	user.login	00000000...	user (00000000...)	{"portal!":"brands"}
8/2/2026, 5:01:09 AM	ad_created	00000000...	ad (00000000...)	{"title!":"Summer Splash Campaign"}
8/1/2026, 5:01:09 AM	review.completed	00000000...	review_session (00000000...)	{"pointsAwarded":120}
7/30/2026, 5:01:09 AM	admin.users_queried	00000000...	admin	{"total":4}
7/28/2026, 5:01:09 AM	redemption.requested	00000000...	redemption (00000000...)	{"amountPoints":500}

Showing 5 of 5 events

PreviousNext

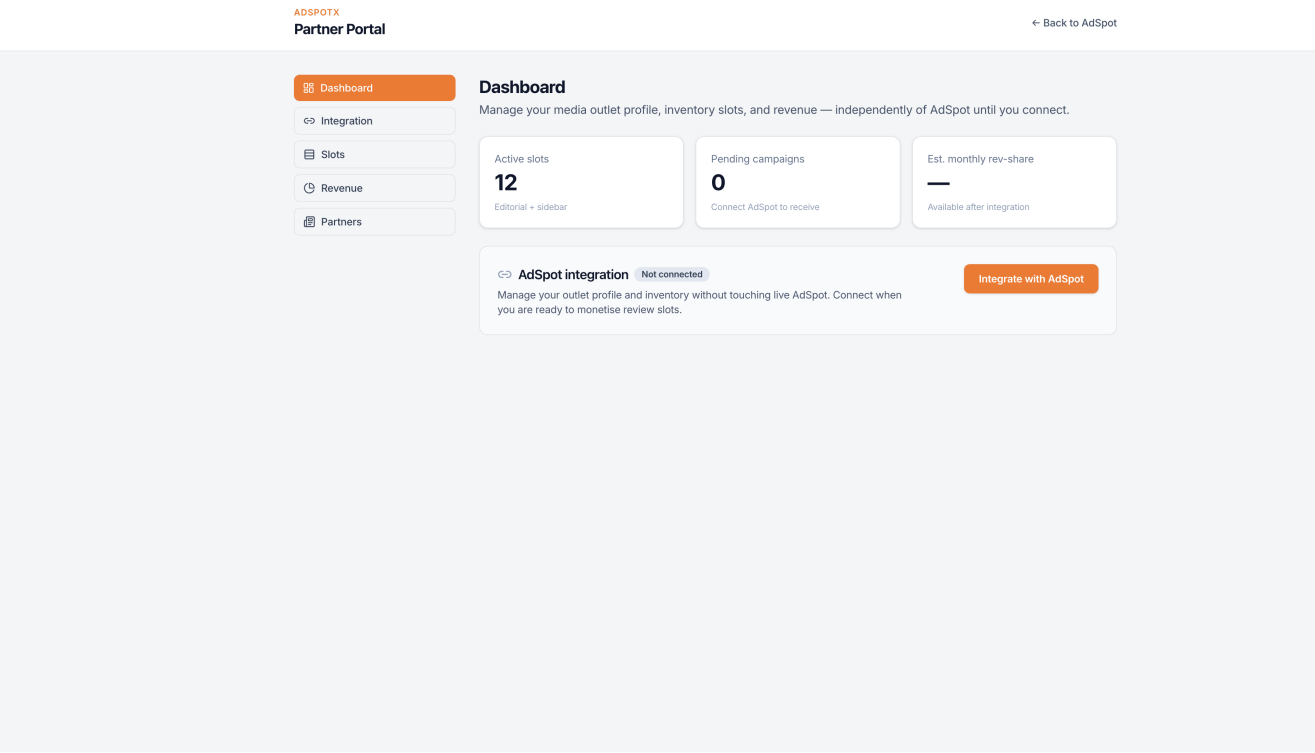
Export CSV

Admin - AdSpotX



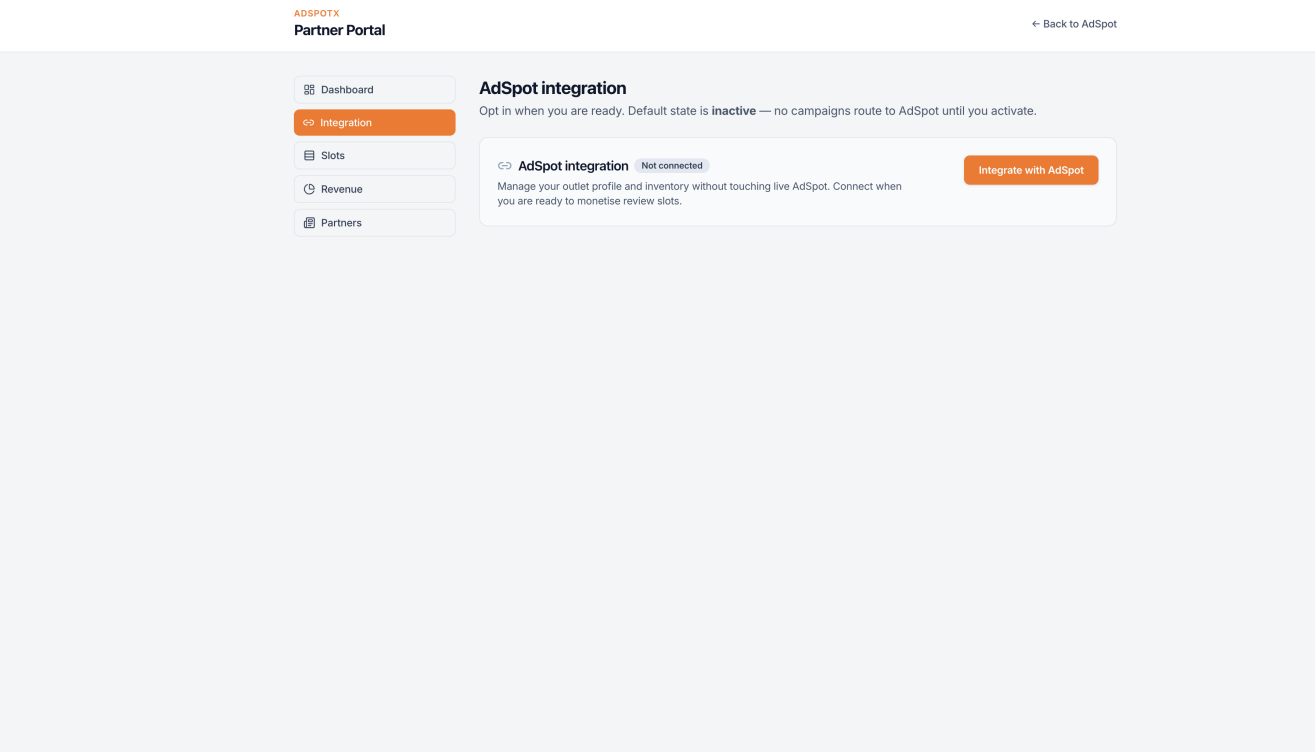
AdSpotX network partner management console

Partner portal



Partner portal dashboard

Partner integration (inactive default)



Partner integration page with Integrate with AdSpot button

7. Market analysis & SWOT

Market context - Africa / Nigeria

Factor	Insight
Digital ad spend	Nigeria digital ad market growing ~15-20% YoY; mobile-first, video-heavy
VAS & rewards	Airtime/data rewards familiar via telco VAS; AdSpot aligns with earned-value apps
Influencer / UGC	Brands shift from vanity impressions to verified attention and comprehension
Publisher pain	Newspapers and digital outlets need programmatic alternatives without heavy ad-tech capex
Youth demographic	18-34 urban Nigerians highly mobile; leaderboard + points drive retention
Payment rails	Paystack/Flutterwave enable NGN wallets and reviewer redemptions

SWOT - AdSpotX

Category	Detail
Strengths	Verified watch-time + survey gating; unified monorepo; one-click partner integration; hostile-audit quality gate; role-separated portals

Weaknesses	Requires Postgres ops; YouTube embed regional limits; early-stage brand pipeline; mock tier needed without DB
Opportunities	Newspaper/network embed via AdSpotX; telco bundle partnerships; FMCG/telco CPE budgets; West/East Africa expansion
Threats	Social platforms' native ad products; fraud in reward apps; FX/NGN volatility; regulatory scrutiny on data/rewards

8. Revenue forecasts (3-year scenarios)

Base currency: NGN millions

Accounting year: August-July, starting FY2026-27

FX assumption: USD 1 = NGN 1,550

Assumptions by scenario

Assumption	Base case	Growth case	Partner-embedded case
Active reviewers (Y3 monthly)	150k	400k	600k
Paying brands (Y3)	80	200	280
Avg brand monthly spend	NGN 2.5M	NGN 4.0M	NGN 3.5M
Network partners (Y3)	5	15	45
Platform take rate	28%	32%	30% (+ partner rev-share)
Partner traffic share (Y3)	10%	25%	55%

Base case P&L (NGN millions)

Line item	Year 1	Year 2	Year 3
Gross brand spend (GMV)	180	720	1,920
Platform net revenue	50	205	538
Partner payouts (net)	8	35	95
Operating costs	42	95	180
EBITDA	0	60	223

Growth case P&L (NGN millions)

Line item	Year 1	Year 2	Year 3
Gross brand spend (GMV)	220	1,200	3,360
Platform net revenue	62	384	1,075
Partner payouts	12	72	210
Operating costs	48	140	320

EBITDA	2	124	505
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Partner-embedded case P&L (NGN millions)

Line item	Year 1	Year 2	Year 3
Gross brand spend (GMV)	200	1,050	2,940
Platform net revenue	56	315	882
Publisher rev-share (60% of partner GMV)	18	95	280
Operating costs	45	120	250
EBITDA	(5)	100	352

Partner-embedded case assumes newspapers/media with existing followership drive reviewer acquisition at lower CAC; platform fee on routed completions remains 10% of gross partner campaign value.

9. Partner flow schematic

Embed flow (newspaper / media partner)

flowchart TD

A[Publisher signs AdSpotX pilot] --> B[Admin creates network_partners record]

B --> C[Partner configures slots in /partners]

C --> D{adspot_linked?}

D -->|false default| E[UI: Not connected - no routing]

E --> F[Partner clicks Integrate with AdSpot]

F --> G[POST /integration/activate]

G --> H[apiKey + embed script issued]

H --> I[Publisher embeds script on site]

I --> J[Reviewers complete campaigns via AdSpot]

J --> K[Webhook: completion verified]

K --> L[Rev-share settlement monthly]

Revenue split (pilot terms)

Brand campaign spend (100%)

%% % 60 % Publisher (network partner)

%% % 30 % Reviewer pool (points / redemptions)

%% % 10 % AdSpot platform (ops, fraud, infrastructure)

Integrate button states

State	UI	API
Inactive	Grey "Not connected"; no embed tag	adspotLinked: false
Active	Green "AdSpot Connected"; copyable script	adspotLinked: true, apiKey, embedScript

10. References & quick start

Related documentation

- docs/ADSPOTX-INTEGRATION.md - Routes, admin workflow
- docs/PARTNER_PORTAL_INTEGRATION.md - Integrate button contract
- docs/ADSPOT_NETWORK_PARTNER_PROGRAM.md - Commercial program
- docs/AWS_RESTAGE.md - AWS staging companion
- docs/ADSPOTX-RELEASE-NOTES-AUG2026.md - v1.2.0 release notes
- STAGING_GUIDANCE.md - Demo accounts and flows

Verify release

```
npx pnpm@9 run build
node scripts/hostile-audit.mjs --mock-only --skip-install
PORT=3001 STATIC_DIR=./app/dist AUDIT_PARTNER MOCK=1 node server/dist/index.mjs
```

Generate this PDF

```
node scripts/render-adspotx-pdf.mjs
# Output: docs/AdSpotX.pdf and ~/Downloads/AdSpotX.pdf
```

End of document - AdSpotX Complete Platform Overview, August 2026.